

HARVARD HACKRARE 2026 | GENOMIC DIAGNOSTICS TRACK

PhenoExtract: Structuring Diagnostic Absence

Mining Negated Clinical Evidence for Rare Disease Diagnosis

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The Problem: Why Rare Disease Diagnosis Fails

Rare disease patients wait 5–7 years for diagnosis.
The bottleneck is not genomics, it is lost phenotypic evidence.



Unstructured Notes

Referral letters, discharge summaries, and clinic notes contain critical phenotypes in free text, never machine-readable.



Lost Negations

"No seizures" vs. "seizures not mentioned" carry opposite diagnostic weight, yet both vanish in standard HPO extraction.

Impact: Tully's Story



Novelty: Modeling Diagnostic Exclusion

Rare disease patients face ~5-year diagnostic delays – often due to missing phenotypic evidence.

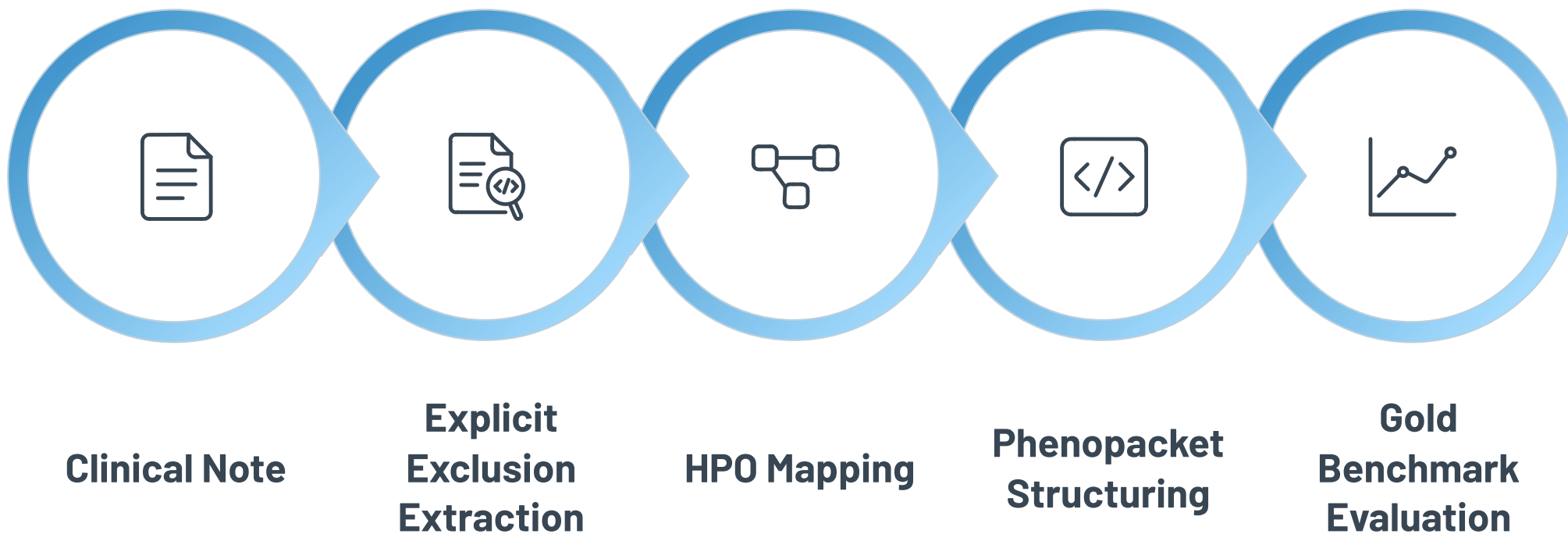
"Patient denies seizures.

No cardiomyopathy.

No respiratory weakness."

"Absence of evidence is still evidence."

PhenoExtract: System Architecture



Feasibility

- **Standards-aligned outputs:** Exclusions are mapped to HPO codes and structured as interoperable phenotype data.
- **Transparent and interpretable:** Outputs include HPO mappings, exclusion status, and negation cues — fully visible in the demo tables.
- **Modular and extensible:** Lightweight extraction pipeline that integrates with existing genomic prioritization tools.

System Demo

30-second walkthrough: Clinical note → Structured phenopacket



Watch Demo Video

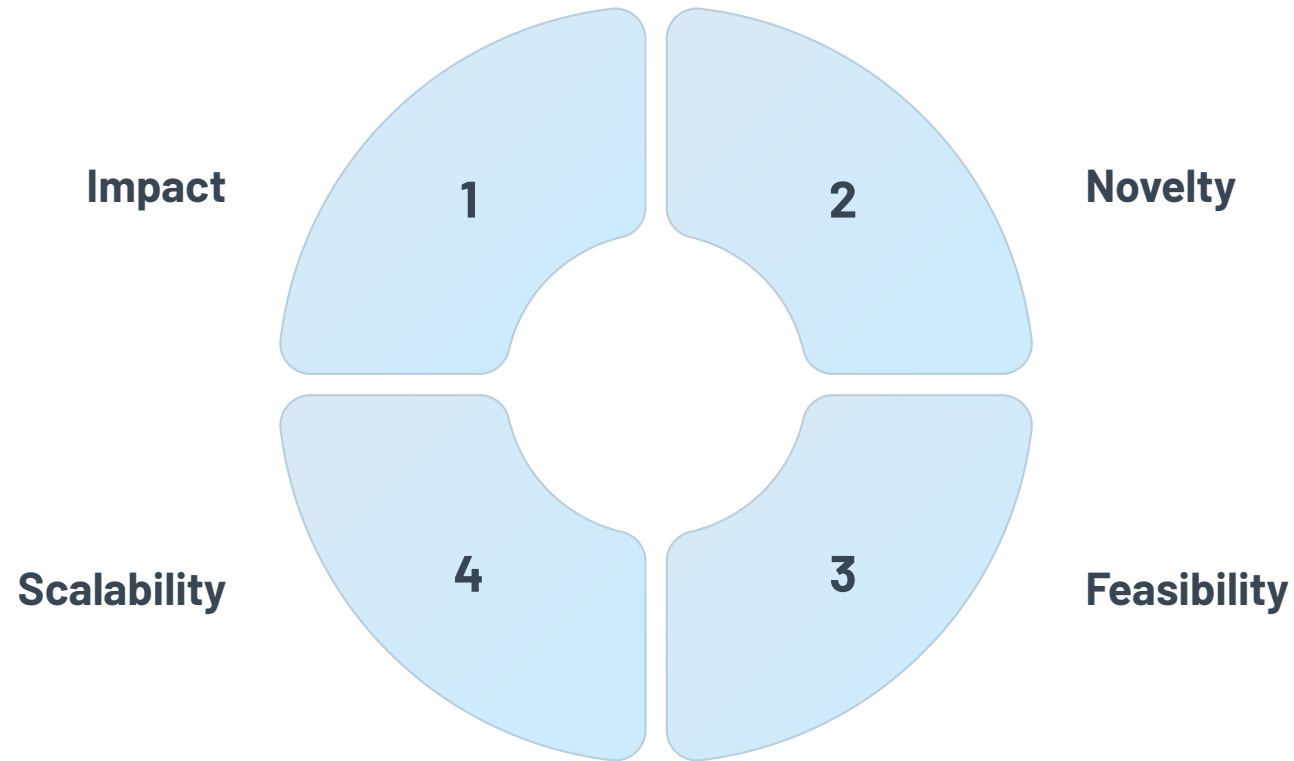
See PhenoExtract in action: raw clinical text transformed into machine-readable diagnostic exclusions ready for genomic analysis.

Limitations

- **Rule-based extraction:** The current system relies on deterministic negation rules, limiting contextual understanding and generalization.
- **Temporal reasoning not modeled:** The system does not yet distinguish between historical, resolved, and ongoing phenotypes.
- **Residual noise:** False positives and missed exclusions remain, particularly in complex or fragmented clinical text.

Meeting every Dimension

Our solution is designed to deliver across all critical success factors.



THANK YOU!